Perception & Decision Making in Intelligent System

Scenario Representation Learning

Group 11
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Action-slot: Visual Action-centric Representations for Multi-label Atomic Activity Recognition in Traffic Scenes

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Introduction

Motivations

- Increased interest in activity recognition in traffic scenes
- Essential for advancing the development of intelligent driving systems
- Multi-label atomic activity recognition, a new task proposed by the OATS dataset for interactive traffic scenario understanding
- Atomic activity: a higher-level semantic motion pattern rooted in the underlying road topology

Limitations

- **Video-level representations** like **3D ConvNet** or **transformers** can’t distinguish individual atomic activities well
- **Object-aware representations** struggles to associate the relationship between the road structures and motions of objects

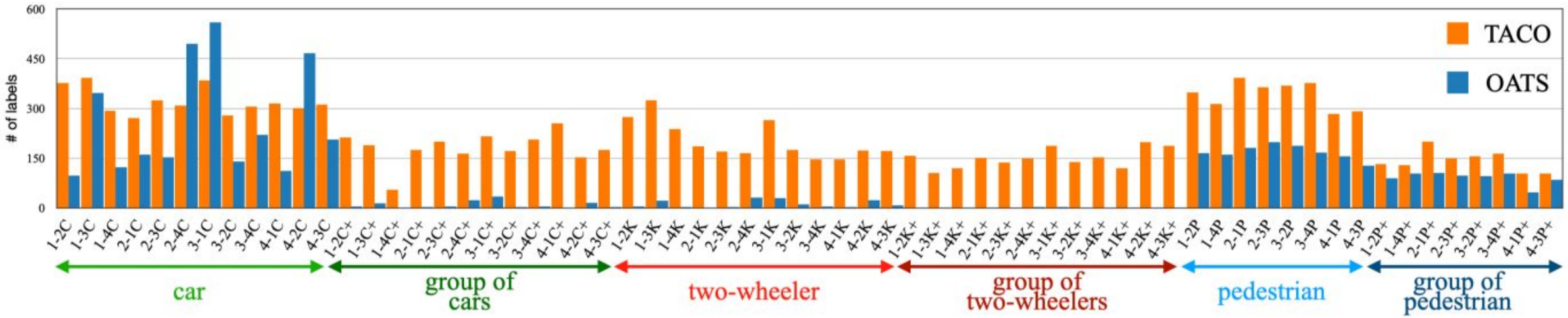
Related work

Traffic Scenario Understanding Datasets

- Previous datasets only labeled high-level actions (e.g. left turn) without location informations
- Inner-City and ROAD dataset solved the problem
- OATS proposed a topology-aware traffic activity description language to unify the action and location labels via the topology-aware pattern description

Traffic Scenario Understanding Datasets

- However, the class distribution of OATS is not balanced
- Constructed the TACO dataset utilizing the CARLA simulator



The distribution of atomic activity classes in our TACO dataset compared to the OATS dataset.

Slot Attention for Representation Learning

- Locatello et al., propose slot attention as a new perspective for learning object-centric representations in an unsupervised manner
- Multi-label atomic activity recognition poses three challenges
  - The slots are permutation invariant
  - Current slot attention architectures are designed for object-centric tasks
  - To enable slot attention in complex scenes, they leverage additional object signals to guide the attention during training

→action-centric slot attention-based framework

Dataset

Dataset-OATS & TACO

Describe how agents interact with traffic road.

- Multi-label action recognition tasks.
  - Balanced data for total 64 classes.
- 6 types of agents:
  1. four-wheeler (C),
  2. Two-wheeler (K),
  3. Pedestrian (P),
  4. Grouped four-wheelers (C+),
  5. Grouped two-wheelers (K+)
  6. Grouped pedestrians (P+).

Z1-Z2:K, Z1-Z3:C, Z4-Z3:C, Z4-Z1:C

2-Z1:K, Z2-Z4:K+, Z3-Z1:K, Z3-Z2:K+, C3-C2:P, C2-C3:P+

	Amount	image resolution	classes
TACO dataset	Testing: 1148 clips; TrainVel: 4030 clips	512x1536	35
OATS dataset	Total: 1026 clips	1200x1920	64

Evaluation Matrics- mean average precision (mAP)

AP_ci = Area under Precision-Recall Curve of class i
Mean (AP of all classes)

Reference: https://sites.google.com/view/road-eccv2024/dataset?authuser=0

Baseline

Baseline

Video-level models

MViT, I3D, X3D, CSN, SlowFast, VideoMAE…

3D ConvNet, self-attention
Learn from video (Consider temporal information) → Predict multi-label output

Object-aware models

ORN, OATS, ARG…

Object Information + Temporal Information → Action Recognition

But…

There may exist objects which don't contribute any information to actions in traffic scenario.

Reference: https://sites.google.com/view/road-eccv2024/dataset?authuser=0

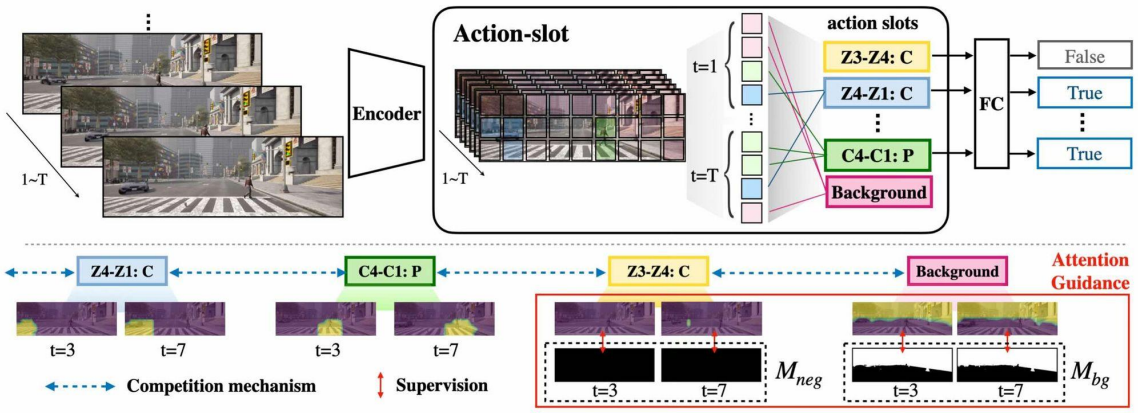
Main Approach

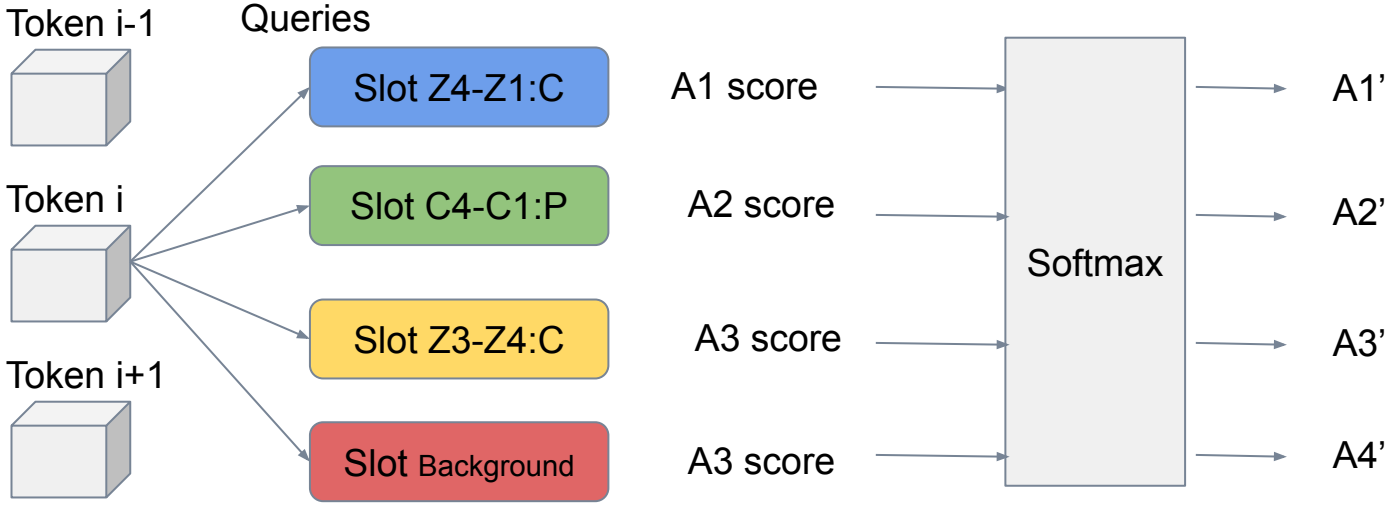
Action Slot

- Attention Mechanism
  Query: Each Slot
  Key : Each tokens

- Each slot correspond to an atomic activity.
  64 slots + BG slot

- Attention Weight: $A = \frac{1}{\sqrt{D}} k(F') \cdot \underbrace{q(S)}$
  Get the softmax along slot-wise
  => Each token will be focused by only 1 slot.
  => **Competition Mechanism**

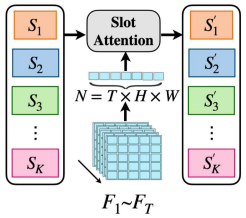




One of values will be significantly large than others.
=>**Dominate Slot**.

Reference: https://sites.google.com/view/road-eccv2024/dataset?authuser=0

Action Slot

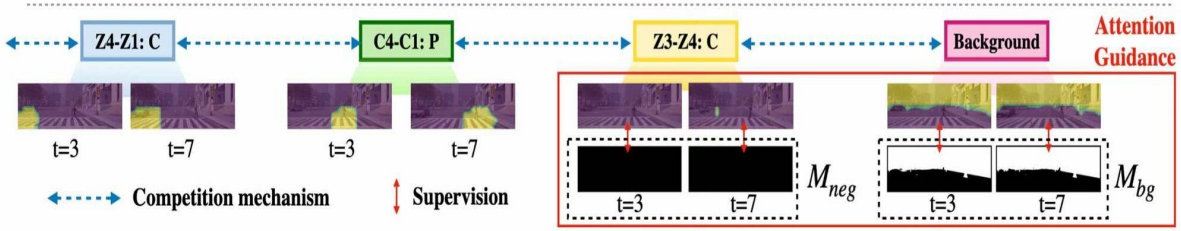


(a) Action-slot: Parallel

- **Objective:**
  $$L_{\texttt{act}} = \sum_{c=1}^{N_{\texttt{cl}}} \mathrm{BCE}(\hat{y}_c, y_c)$$

- **Parallel updating**
  Consider temporal dimension at the same time.

- **Background slot & attention guidance**
  Add BG guidance for BG slot.
  $$L_{\texttt{bg}} = \mathrm{BCE}(\bar{A}_{\texttt{bg}}, M_{\texttt{bg}})$$



- **Regularization for action slots.**
  Avoid those activities not present in a video from attending to any regions.
  $$L_{\texttt{neg}} = \sum_{\{c|y_c=0\}} \mathrm{BCE}(\bar{A}_c, M_{\texttt{neg}})$$

Total Loss: Lact + Lbg+ Lneg

Reference: https://sites.google.com/view/road-eccv2024/dataset?authuser=0

Results Analysis

Results Anaysis

With better results:

1. X3D– Kinetics-400
2. Slot-based models (proposed method)

Video-level model

Object-aware model

Slot-based models

OATS dataset

Method	Backbone	Seq	Pretrain	S1	S2	S3	mAP
CSN [3, 71]	ResNet152	32	IG65M	12.1	12.6	12.9	12.5
TPN [3, 78]	ResNet50	32	ImageNet	11.6	13.3	12.9	12.6
SlowOnly [3, 28]	ResNet50	32	ImageNet	11.2	14.7	12.9	13.0
SlowFast [3, 28]	ResNet50	32	None	10.8	15.1	14.5	13.5
I3D (NL) [3, 73]	ResNet50	32	ImageNet	11.9	15.5	14.0	13.8
I3D [3, 12]	ResNet50	32	ImageNet	11.8	14.3	16.8	14.3
ORN [3, 6]	ResNet50	32	ImageNet	16.8	13.4	18.1	16.1
ARG [3, 74]	Inceptionv3	32	ImageNet	20.2	21.3	19.3	20.3
OATS [3]	Inceptionv3	32	ImageNet	24.3	28.6	27.2	26.7
SAVi‡ [24, 41]	ResNet50	32	ImageNet	21.1	22.4	22.5	22.0
MO‡ [5]	ResNet50	32	ImageNet	14.3	15.6	18.2	16.0
Slot-VPS‡ [81]	ResNet50	32	ImageNet	15.7	17.8	17.2	16.9
Action-slot (Ours)	ResNet50	32	ImageNet	**26.6**	**28.6**	**30.8**	**28.6**
I3D [12]	ResNet50	8	Kinetics-400	21.7	24.6	24.4	23.6
X3D [27]	N/A	16	Kinetics-400	30.4	33.2	30.6	31.4
CSN [71]	ResNet101	32	Kinetics-400	43.1	47.1	44.3	44.8
SlowFast [28]	ResNet50	16	Kinetics-400	36.1	36.6	34.2	35.6
ORN [6]	X3D	16	Kinetics-400	19.3	22.5	23.6	21.8
ARG [74]	X3D	16	Kinetics-400	24.8	25.9	29.3	26.7
SAVi‡ [24, 41]	X3D	16	Kinetics-400	19.0	22.1	21.6	20.9
MO‡ [5]	X3D	16	Kinetics-400	25.3	25.0	24.2	24.8
Slot-VPS‡ [81]	X3D	16	Kinetics-400	24.7	24.6	25.5	24.9
Action-slot (Ours)	X3D	16	Kinetics-400	**48.1**	**47.7**	**48.8**	**48.2**

TACO dataset

Method	Para. (M)	Seq	C	K	P	C+	K+	P+	mAP
I3D [12]	27.3	8	27.3	19.4	30.5	34.6	33.6	34.8	29.7
X3D [27]	3.0	16	37.5	20.3	34.6	56.3	51.5	38.8	38.3
CSN [71]	21.4	32	43.5	35.5	43.0	52.5	46.1	43.4	44.0
SlowFast [28]	33.7	16	33.9	19.7	36.7	39.9	42.6	41.0	35.2
MViT [26]	36.6	16	21.4	13.8	26.3	43.7	30.0	33.8	27.9
VideoMAE [26]	57.9	16	30.6	18.6	27.1	51.6	33.0	37.4	33.1
ORN [6]	4.8	16	25.5	15.8	24.6	31.6	19.8	13.9	22.2
ARG [74]	12.2	16	27.8	15.0	27.2	35.6	13.7	20.0	23.2
SAVi‡ [24, 41]	2.3	16	19.2	13.6	27.5	23.3	25.9	35.9	23.3
MO‡ [5]	2.3	16	34.2	24.9	39.2	38.3	39.4	37.7	35.3
Slot-VPS‡ [81]	3.5	16	31.9	21.3	32.0	51.9	44.7	31.3	36.0
Action-slot (Ours)	**2.3**	16	**48.1**	**41.2**	**49.2**	**70.1**	**62.6**	**52.8**	**54.4**

Reference: https://sites.google.com/view/road-eccv2024/dataset?authuser=0

Thanks